

Binary RR Lyrae in the LMC from the light-travel-time effect

What we measure (the $O-C$ diagram), mock illustrations, and the pipeline

Project `rrl_lmc_binary` — working note

1 October 2026, pipeline v3 (code:

https://github.com/vasilybelokurov/rrl_lmc_binary)

Abstract

An RR Lyrae star pulsates with a period of about half a day and a very stable, sharply shaped light curve, so it can be used as a clock. If it orbits a companion, its distance from us changes periodically and the pulsation signal arrives earlier or later by the light-travel time across the orbit — minutes to tens of minutes for companions of $0.1\text{--}1\,M_\odot$ on 3–30-yr orbits. We measure this delay once per observing season for 17,488 LMC RRab stars with OGLE-II/III/IV photometry (1997–2016) and, for 6,612 of them, MACHO B and R photometry (1992–1999). The resulting time series of delays is the $O-C$ diagram. This note explains the measurement in words and in equations, illustrates it with mock data, and documents the pipeline and its tests (sections 1 to 4). It then gives the results: the candidate selection (section 5; 75 candidates, 11 in the cleanest tier); the calibration on simulations (section 6), which shows that the real stars carry red, mostly intrinsic timing noise that accounts for the number of candidates and gives an upper limit $f < 2.7\%$ on the fraction with $0.4\text{--}1.5\,M_\odot$ companions at $P_{\text{orb}} = 1\text{--}10\text{ kd}$; a comparison with the Galactic-bulge candidates and a Gaia test (section 7); and frozen predictions for 2016–2026 that post-2016 OGLE-IV photometry would test decisively (section 8).

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1 What we measure, in simple words

1. **The clock.** An RRab star brightens steeply (within about an hour) and fades slowly, repeating every $P \approx 0.5\text{--}0.7$ d. The shape repeats almost exactly, so the times of the steep rise are sharp time markers.
2. **The prediction.** From the catalogue period and one reference epoch we predict when the rise should happen in any later year: the “calculated” time C .
3. **The observation.** In each observing season we find the time shift that best lines up the observed light curve with the star’s own template: the “observed” time O . The difference $O - C$ is the *delay* of that season.
4. **What changes the delay.** (i) A slightly wrong catalogue period makes $O - C$ grow linearly with time. (ii) A slow, steady change of the period (stellar evolution) bends $O - C$ into a parabola. (iii) An orbiting companion moves the RR Lyrae towards and away from us; the light has a little further or a little less distance to travel, so $O - C$ oscillates with the orbital period (figure 1). This is the light-travel-time effect (LTTE). (iv) The star itself can change its period irregularly (jumps, wander, Blazhko modulation); this also moves $O - C$ and is the main confusion.

5. **Why it works at the LMC.** The delay depends only on the size of the orbit, not on the distance to the star. A $0.6 M_{\odot}$ companion on a 3000-d orbit gives ± 17.5 min everywhere in the Universe. The price of the LMC distance is photometric noise: our stars are at $I \approx 19$, so one season's delay is measured to about 4 min (median).
6. **What we look for.** A delay that oscillates smoothly and repeats, with the same amplitude in every band and survey, on top of an ordinary smooth period change; and, statistically, how many such stars there are.

2 What we measure, in equations

2.1 Light curve, ephemeris and delay

Let $T(\phi)$ be the star's mean light curve (template) as a function of pulsation phase $\phi \in [0, 1)$, written as a Fourier series of order K (we use $K = 8$),

$$T(\phi) = \sum_{k=1}^K [a_k \cos(2\pi k\phi) + b_k \sin(2\pi k\phi)]. \quad (1)$$

With a reference period P and epoch T_0 (the OGLE-IV catalogue values), the observed magnitude is modelled as

$$m(t) = z_s + \delta m_j + \alpha_j T\left(\frac{t - \tau_j - T_0}{P}\right) + \varepsilon, \quad (2)$$

where s labels the survey segment (OGLE-II, -III, -IV, MACHO; separate zero points z_s), j the observing season, τ_j the season's delay, δm_j a mean-magnitude offset and α_j an amplitude factor (both nuisance parameters; α_j is also a Blazhko diagnostic), and ε the photometric noise. A single τ_j shifts all harmonics together: this is what a change in light-travel time does. The set $\{\tau_j\}$ versus time is the $O-C$ diagram; $\tau > 0$ means the pulsation arrives late.

2.2 Ordinary period change: the quadratic

If the period changes slowly and steadily, $P(t) = P_0 + \dot{P}t$, the n -th maximum occurs at $T_n = T'_0 + nP_0 + \frac{1}{2}\dot{P}P_0n^2$, and with $n \simeq t/P_0$

$$\tau(t) = c_0 + c_1 t + c_2 t^2, \quad c_1 = \frac{P_0 - P}{P}, \quad c_2 = \frac{\dot{P}}{2P}. \quad (3)$$

c_0 is an uninteresting constant (epoch and template-phase convention), c_1 a period error, c_2 the rate of period change. For an evolutionary $\dot{P} = 0.1 \text{ d Myr}^{-1}$ ($2.7 \times 10^{-10} \text{ d d}^{-1}$) and $P = 0.55 \text{ d}$, $c_2 = 2.5 \times 10^{-10} \text{ d}^{-1}$, i.e. $\tau \approx 2.5 \times 10^{-10} \times (4000)^2 \text{ d} = 0.004 \text{ d} \approx 350 \text{ s}$ at 4000 d from the reference epoch: comparable to the orbital signals, so the quadratic is always part of the model.

2.3 Light-travel-time effect

If the RR Lyrae (mass M_1) orbits a companion (M_2), the delay is its line-of-sight position divided by c [1]:

$$\tau_{\text{LTTE}}(t) = \frac{a_1 \sin i}{c} \frac{1 - e^2}{1 + e \cos \nu(t)} \sin(\nu(t) + \omega), \quad (4)$$

with true anomaly ν from Kepler’s equation, eccentricity e , argument of periastron ω , and

$$a_1 = \left[\frac{G(M_1 + M_2)P_{\text{orb}}^2}{4\pi^2} \right]^{1/3} \frac{M_2}{M_1 + M_2}. \quad (5)$$

From the fitted P_{orb} and semi-amplitude $A \equiv a_1 \sin i/c$ one obtains the mass function

$$f(M_2) = \frac{4\pi^2(cA)^3}{GP_{\text{orb}}^2} = \frac{(M_2 \sin i)^3}{(M_1 + M_2)^2}, \quad (6)$$

and the primary’s radial-velocity semi-amplitude $K_1 = 2\pi cA/[P_{\text{orb}}\sqrt{1-e^2}]$. The full $O-C$ model is $\tau(t) = c_0 + c_1t + c_2t^2 + \tau_{\text{LTTE}}(t)$.

Table 1: LTTE semi-amplitude $A = a_1/c$ for $M_1 = 0.65 M_\odot$, circular, edge-on (`rrlbin.ltte.a1sini_over_c`).

P_{orb} [d]	$M_2 = 0.6 M_\odot$	$0.2 M_\odot$	$0.07 M_\odot$
1000	505 s	218 s	85 s
3000	1050 s (17.5 min)	453 s	177 s

2.4 Measuring one season’s delay

For fixed template (a_k, b_k, z_s) , each season’s $(\tau_j, \delta m_j, \alpha_j)$ is fitted by weighted least squares (Gauss–Newton after a coarse search over one full cycle). Its Fisher (Cramér–Rao) error is

$$\sigma_{\tau_j}^{-2} \simeq \sum_{i \in j} \frac{1}{\sigma_i^2} \left(\frac{\partial m}{\partial t} \right)_i^2 = \sum_{i \in j} \frac{\alpha_j^2 T'(\phi_i)^2}{P^2 \sigma_i^2}, \quad (7)$$

scaled by $\sqrt{\max(\chi_\nu^2, 1)}$. The steep rising branch dominates the sum. Template and delays are iterated to convergence. Because a constant delay is degenerate with the template phase, we fix the gauge so that the template’s fundamental harmonic has phase zero.

2.5 Detection statistic D

On the season delays $\{t_j, \tau_j, \sigma_j\}$ we compare two Gaussian models with variance $\sigma_j^2 + s^2$ (extra jitter s fitted in each): H_0 : [equation \(3\)](#) (+ MACHO B and R offsets with band-lag priors, [section 4.5](#)); H_1 : $H_0 + A_s \sin(2\pi t/P_{\text{orb}}) + A_c \cos(2\pi t/P_{\text{orb}})$.

$$D = 2 \max_{P_{\text{orb}}} [\ln \mathcal{L}_1(P_{\text{orb}}) - \ln \mathcal{L}_0]. \quad (8)$$

At fixed P_{orb} and for white noise $D \sim \chi_2^2$; for N seasons of error σ and amplitude A , $D \approx NA^2/(2\sigma^2)$. D tests an orbit against a *smooth* period change plus white noise; it does not by itself reject irregular intrinsic period changes ([section 4.8](#)).

3 Mock-data illustrations

All figures in this section are synthetic (`scripts/plot_writeup_mock.py`); the template is a typical OGLE I -band RRab template (first-harmonic amplitude 0.155 mag), and the orbit has $M_1 = 0.65$, $M_2 = 0.6 M_\odot$, $P_{\text{orb}} = 3000$ d, $e = 0.3$, $i = 90^\circ$.

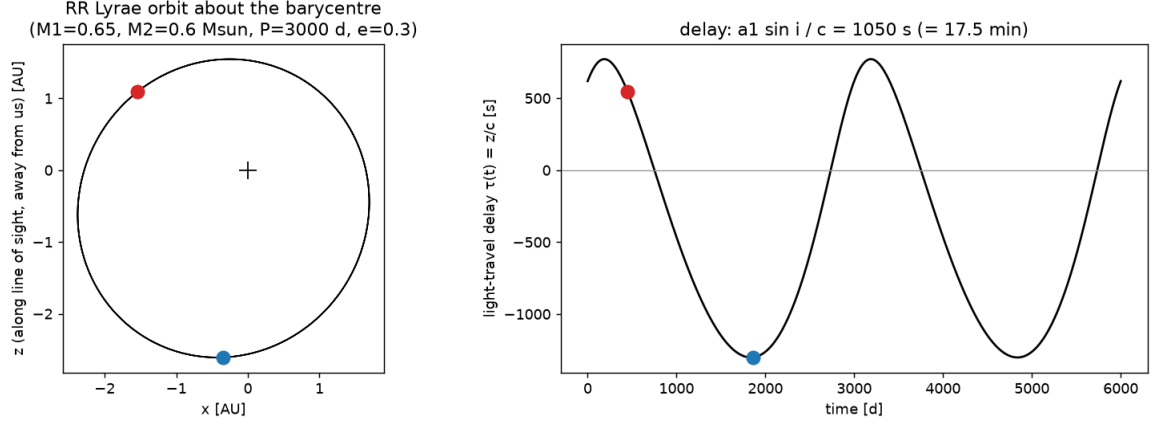


Figure 1: Left: the RR Lyrae's orbit about the barycentre (+), with the observer below. Right: the resulting delay $\tau(t) = z/c$ (equation (4)); the non-sinusoidal shape comes from $e = 0.3$. Coloured dots mark the same two epochs in both panels: far side (red, late) and near side (blue, early).

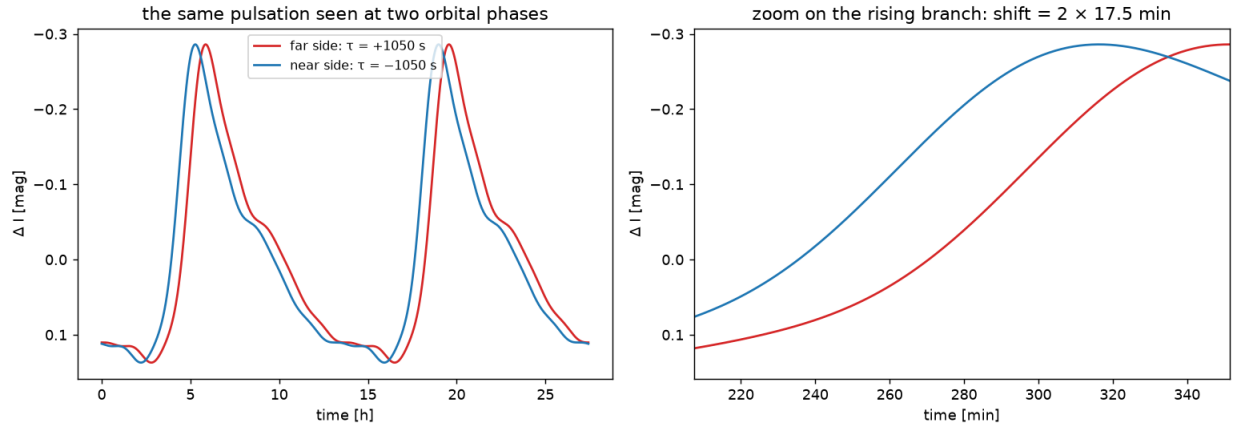


Figure 2: The same pulsation observed at the two orbital phases of figure 1: the light curve is not changed in shape or amplitude, only shifted in time. Right: zoom on the rising branch, where the shift (2×17.5 min) is most easily measured.

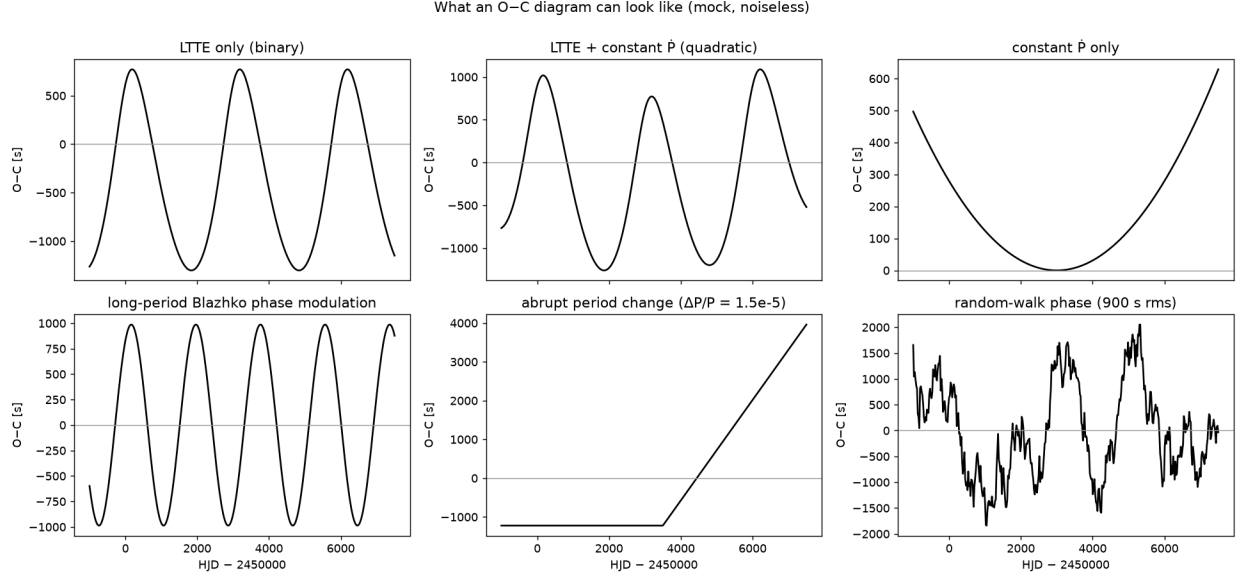


Figure 3: Noiseless mock $O-C$ diagrams over the 1992–2016 baseline. Top: binary (LTTE), binary plus constant $\dot{P}$, constant $\dot{P}$ only. Bottom: the intrinsic nuisances — long-period Blazhko phase modulation (periodic, like an orbit, but accompanied by amplitude changes), an abrupt period change (a broken line), and random-walk phase wander. Over one or two cycles all three can resemble an orbit.

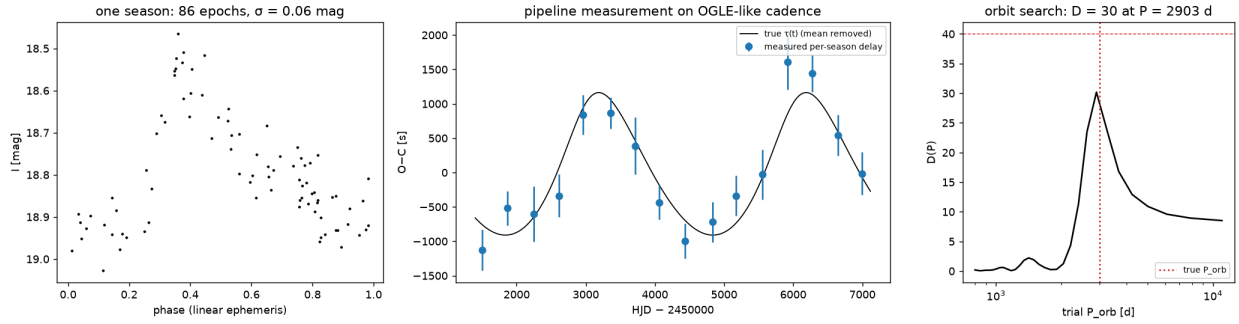


Figure 4: The pipeline on a mock star with an OGLE-III+IV-like cadence (16 seasons, 40–110 epochs each, $\sigma = 0.06$ mag). Left: one season folded on the linear ephemeris. Middle: measured season delays (median error 297 s) against the true delay. Right: the orbit search $D(P_{\text{orb}})$ (equation (8)) recovers $P = 2903$ d (true 3000 d) with $D = 30$ — a 1050-s signal is a firm but not overwhelming detection at LMC photometric precision.

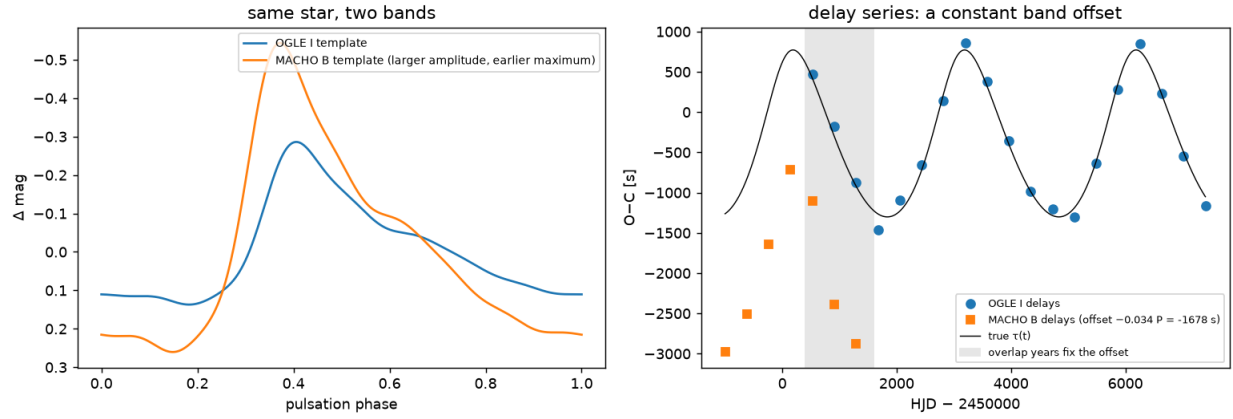


Figure 5: Two surveys in two bands. Left: MACHO *B* and OGLE *I* templates of the same star differ in amplitude and the maximum occurs 0.034 cycles earlier in *B* (the measured mean lag, [section 4.5](#)). Right: hence the MACHO delay series is offset by a constant from the OGLE series; the years observed by both surveys (grey) fix the offset.

4 The pipeline

4.1 Data

- **OGLE-IV** LMC RR Lyrae collection [3]: 41,471 stars; *I*-band light curves for 41,209, epochs $\text{HJD}' = 5260\text{--}8924$, where $\text{HJD}' = \text{HJD} - 2450000$ (for 95% of the stars the public data end at $\text{HJD}' \approx 7516$, mid-2016). <https://www.astrouw.edu.pl/ogle/ogle4/OCVS/lmc/rrlyr/>
- **OGLE-III** [2]: 24,906 light curves, $\text{HJD}' = 2166\text{--}4540$ (2001–2008); the 2024 re-issue of the files includes **OGLE-II** epochs (from $\text{HJD}' = 447$, 1997) for 7,784 stars. Same star numbering as OGLE-IV (24,818 matches, median separation $0.05''$). <https://www.astrouw.edu.pl/ogle/ogle3/OIII-CVS/lmc/rrlyr/>
- **MACHO** *B* and *R* photometry (1992–1999) via the NCI TAP service (<https://machotap.svo.nci.org.au/ncitap/tap>), queried per tile for the RR Lyrae only (MACHO IDs from the OGLE-III cross-identification): 8,406 of 8,557 requested stars returned. Both bands are used.
- **Sample**: 17,488 RRab with a successful OGLE *I* timing fit; 6,612 of them with MACHO (6,599 with both *B* and *R*), 10,876 OGLE only.

4.2 Time system

OGLE times are HJD (UTC). MACHO `dateobs` is MJD and is converted to HJD for each star (observer at Mount Stromlo); towards the LMC, near the south ecliptic pole, the heliocentric correction is only ± 40 s (tested). MACHO time stamps refer to the start of the exposure (+150 s to mid-exposure; Michalska & Pigulski 12); a constant offset of this kind is absorbed by the per-band MACHO offset (section 4.5).

4.3 Seasons

A season is one observing year, with the boundary at $\text{HJD}' \bmod 365.25 = 245$ (the middle of the OGLE seasonal gap). MACHO observed the LMC nearly year-round, so gap-based seasons would merge years. Seasons with fewer than 15 epochs are dropped.

4.4 Timing fit per star and band

equation (2) with $K = 8$, one template per band (OGLE *I*, MACHO *B*, MACHO *R*), separate zero points per survey segment, per-season ($\tau_j, \delta m_j, \alpha_j$), 4σ clipping, template $\leftrightarrow$ delay iterations until the mask is stable and $\max |\Delta\tau| < 10^{-7}$ d, the fundamental-phase gauge, and errors from equation (7) scaled by $\sqrt{\max(\chi_\nu^2, 1)}$. Each season also yields a *harmonic coherence*: the delay measured from the fundamental minus that measured from the higher harmonics. For a pure time shift (LTTE) it is zero within errors; a change of light-curve shape makes it non-zero. Delays are defined modulo P ; they are unwrapped between consecutive seasons, and a slip is repaired only if the repair reduces the jumps to both adjacent seasons (a more aggressive rule made noisy stars worse). *Measured precision*: median season error 189 s for MACHO stars and 270 s for OGLE-only stars (combined over bands), strongly amplitude-dependent.

4.5 Tying MACHO to OGLE

Band-dependent template shapes leave a constant offset Δ between each MACHO delay series and the OGLE *I* series (figure 5). The same behaviour is found between OGLE’s own *V* and *I*, which

share one time system: the offset is a physical band lag, not a clock error. Calibrated on the full sample (`scripts/calibrate_band_lag.py`):

$$\Delta_B = -6648 P + 1966 \text{ s } (\sigma_{\text{int}} = 293 \text{ s}, N = 1582), \quad \Delta_R = -2656 P + 718 \text{ s } (\sigma_{\text{int}} = 218 \text{ s}, N = 1552), \quad (9)$$

with P in days (-0.036 and -0.016 cycles). In the $O-C$ fits each MACHO band has its own offset with this relation as a Gaussian prior (pseudo-observations, identical under H_0 and H_1). About 13% of MACHO stars had a band series a whole pulsation cycle away from the prior after unwrapping; these are shifted by the integer number of cycles that best matches the prior (`oc.align_bands`). Simulated stars draw their MACHO offsets from the same relation and scatter.

4.6 Common-mode timing correction

Stacking all stars' $O-C$ residuals by (band, year) reveals a common pattern. It is solved iteratively as a clipped mean, with the directions that are degenerate with each star's own model (a common quadratic and per-band constants) projected out; candidates are excluded. The MACHO B and R patterns agree within a few seconds (1992–1999, B : $-37, +26, +34, +39, +56, +53, -60, -110$ s; R : $-38, +24, +32, +41, +55, +49, -63, -99$ s), so this is a timing systematic of the MACHO data; the OGLE pattern spans -66 to $+46$ s. A null test (white-noise delays at the real epochs) is consistent with zero. The correction is subtracted from every star.

4.7 Orbit search and the annual-sampling limit

D (equation (8)) is maximized over trial periods uniform in frequency, from 400 d to twice the baseline; H_1 is fitted both with a circular orbit (one harmonic) and with two harmonics (eccentric orbits). Season delays sample the $O-C$ about once per year, but the epochs of the seasons scatter by 40 d rms within the year; the spectral window at 1/yr is 0.85, not 1, and injected orbits from 300 d upwards are recovered at the true period in 86–100% of cases (`scripts/alias_test.py`). The real short-period limit is the averaging of the orbit over a season of ≈ 240 d, which reduces the amplitude by $|\text{sinc}(\pi 240/P_{\text{orb}})| = 0.23, 0.50, 0.66$ at 300, 400, 500 d; hence the grid starts at 400 d. An alias flag records the difference in D between the best period and its $\pm 1/\text{yr}$ aliases.

4.8 Noise models, nuisances and discriminators

Simulations use each star's real cadence and errors and its own fitted templates (`src/rrlbin/simulate.py`); one delay realization is shared by all bands. Classes: white noise (null); Blazhko (coherent phase and amplitude modulation, $P_B = 20\text{--}3000$ d); abrupt period changes, small and large; random-walk phase, small and large; LTTE (log-uniform $P_{\text{orb}} = 400\text{--}10^4$ d, $M_2 = 0.05\text{--}1.5 M_{\odot}$, isotropic, half eccentric); and an **empirical** class that reproduces the real stars' excess timing noise (below).

The real excess timing noise is red. Real stars scatter about H_0 far more than the photometric errors allow (χ^2_{ν} p50/p90 $\approx 2/30$, against $\approx 1/2$ for white-noise simulations, which do reproduce the real per-season errors to within 2–3%). The excess is common to the bands: same-season residuals correlate with Spearman $\rho = 0.58$ between MACHO B and R and $\rho = 0.37$ between MACHO and OGLE (nulls: ≈ 0). It is mostly intrinsic to the stars. It is also *smooth*: per survey, the scatter about a quadratic is much smaller than over the joint baseline, whereas a Brownian random walk overpredicts it by a factor $\sim 7\text{--}10$. We model it as a Gaussian process (GP) wander plus white jitter,

$$y_j = \mathbf{x}_j^T \boldsymbol{\beta} + w(t_j) + \epsilon_j, \quad \langle w(t)w(t') \rangle = A_w^2 \exp\left[-\frac{(t-t')^2}{2\ell^2}\right], \quad \epsilon_j \sim \mathcal{N}(0, \sigma_j^2 + s^2), \quad (10)$$

fitted by restricted maximum likelihood (REML; ML biases A_w low) on a grid $\ell \in \{700, 1500, 3000, 6000\}$ d (`oc.gp_null`). Real fits give A_w/σ p50/p75/p90 = 2.1/11/22 (MACHO stars) and 1.5/7.8/22 (OGLE-only); $\ell = 700$ d is the most common. The **empirical** class draws $(s/\sigma, A_w/\sigma, \ell)$ from the real fits (candidates excluded) and splits the white jitter 64% common to all bands and 36% per instrument (MACHO $B+R$ shared), which reproduces the measured band correlations ($\rho_{BR} = 0.63$, $\rho_{MO} = 0.37$). The per-band χ^2_ν of the empirical simulations matches the real stars within a factor 1.5–2.5, with the simulations slightly noisier: conservative for false alarms.

Discriminators (computed identically for data and simulations):

- amplitude constancy $\chi^2_\nu(\alpha_j) < 2$, pooled over bands (Blazhko veto; LTTE does not change amplitude);
- harmonic coherence χ^2_ν (shape changes; LTTE shifts all harmonics equally);
- at least 1.5 orbital cycles within the baseline;
- a physical ceiling: A below a_1/c for $M_2 = 2 M_\odot$ at the fitted period;
- for MACHO stars, a predictive test: the orbit period and parameters are found from the seasons after $\text{HJD}' = 450$ only (unwrapping and band alignment also done on these training seasons), then the held-out 1992–1996 MACHO delays are predicted; the score is the log-likelihood gain of the orbit prediction over the H_0 prediction.

4.9 Code and tests

Package `rrlbin`: `io`, `timing` (per-season fits), `pipeline` (Level 1: star $\rightarrow$ delay series), `oc` (Level 2: unwrapping, band alignment, common mode, orbit search, GP noise model, predictive test; shared by data and simulations), `ltte`, `simulate`, `kepler_fit` (Keplerian fits), `predict` (kriging predictions), `mixture`. 40 unit tests (`pytest`, all pass), including recovery of noiseless delays, gauge invariance, pull distributions, the Kepler solver, the null calibration of D , the alias test, leakage-free predictive scores, band alignment, GP recovery (A_w 950 vs 900 s injected at $\ell = 1500$ d), Keplerian recovery of $e = 0, 0.4, 0.7$ orbits and the kriging limits. Two bugs found by tests in v3 (amplitudes 19% low from a parameter-name collision; prediction leakage) are fixed with regression tests. Long runs use a chunked, resumable runner with a manifest check. Full log: `JOURNAL.md`.

5 Selection of candidates

Figure 6 summarizes the v3 $O-C$ fits of all 17,488 R Rab (after the common-mode correction, with band-lag priors, on the 400-d period grid) together with 26,982 simulations run through the same pipeline on the real cadences of 1,500 MACHO+OGLE stars (`results/inject/stats_v3_macho.parquet`; per star 2 null, 2 Blazhko, 1 small jump, 1 small random walk, 2 large jumps, 2 large random walks, 4 empirical-noise and 4 LTTE). A second campaign on 800 OGLE-only stars (14,400 simulations) calibrates the OGLE-only sample. All numbers below are from `scripts/selection_numbers.py` (`results/real/selection_numbers_v3.json`); the figure from `scripts/plot_summary_stats.py`.

5.1 The panels

(a) Detection statistic.

Distribution of D (equation (8)) for real stars (grey) and four simulated classes. White-noise nulls (black; a constant $\dot{P}$ and photometric noise only) never exceed $D = 23.0$ in 2,998 simulations; their

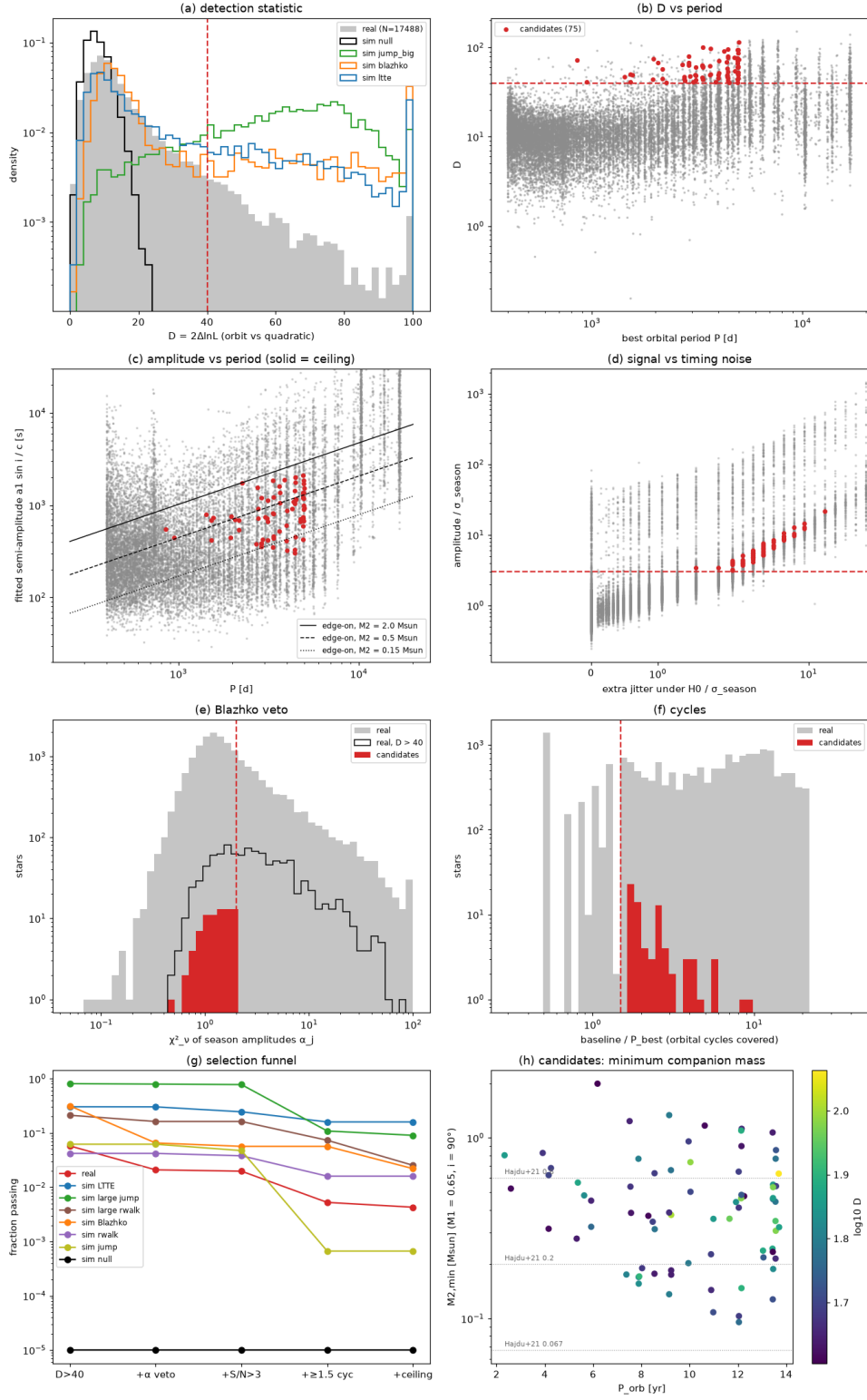


Figure 6: Summary statistics of the v3 $O-C$ fits and the candidate selection. Grey: all real R Rab; red: the 75 candidates; coloured lines: simulated classes (MACHO campaign). Panels are described in [section 5.1](#); the cuts in [table 2](#).

per-star false-alarm thresholds are $D = 13.8$ (5%), 16.9 (1%) and 20.5 (0.1%). Real stars have a long tail far beyond the nulls: 1,009 (5.8%) have $D > 40$, where white noise predicts none. Simulated orbits (blue), Blazhko modulation (orange) and large abrupt period changes (green) all populate that tail, so D alone cannot separate orbits from intrinsic period changes. (The empirical class, which reproduces the real tail, is shown in [figure 7a.](#))

(b) D versus best period.

Each star’s D at the period that maximizes it. The vertical banding is the discreteness of the frequency grid (nearly all MACHO stars share one baseline and hence the same grid points). Candidates (red; 81% of them) concentrate at 2,500–5,000 d, where a 24-yr baseline covers about 2–4 cycles and the search is most sensitive.

(c) Amplitude versus period.

Fitted semi-amplitude $A = a_1 \sin i/c$ with the edge-on LTTE amplitude for $M_2 = 0.15, 0.5$ and $2 M_\odot$ ($M_1 = 0.65 M_\odot$). The solid line is the physical ceiling used in cut 5. The cloud above it at long periods consists of stars whose $O-C$ changes by a large fraction of a pulsation cycle: intrinsic period changes far larger than any companion of $\leq 2 M_\odot$ can produce. Candidates lie between the 0.15 and $2 M_\odot$ curves, 37% above the $0.5 M_\odot$ curve.

(d) Signal versus timing noise.

Orbit amplitude and extra jitter (fitted under H_0), both in units of the median season error. Stars with no extra jitter under H_0 form the column at zero. A real orbit not modelled under H_0 is absorbed as jitter, so strong orbits move up and to the right together; the discrete columns are the jitter grid. Cut 3 (dashed) removes signals that are not resolved season by season.

(e) Blazhko veto.

χ_ν^2 of the per-season amplitude factors α_j about their mean. A light-travel delay shifts the light curve without changing its amplitude ($\chi_\nu^2 \approx 1$), whereas Blazhko modulation changes it: the median χ_ν^2 is 0.97 for simulated orbits and 0.98 for nulls, but 1.9 for simulated Blazhko stars and 1.46 for real stars. The black histogram (real, $D > 40$) is shifted to large χ_ν^2 : most high- D real stars change amplitude from season to season.

(f) Cycles covered.

Baseline divided by the best period. At fewer than 1.5 cycles a sinusoid is not distinguishable from a smooth curve (cubic trend, broken line, smooth wander).

(g) Selection funnel.

Fraction of each class passing the cuts applied in sequence. Real stars (red) fall from 5.8% to 0.43%; simulated orbits (blue) from 30% to 16%; large abrupt period changes (green) from 81% to 9%; white-noise nulls pass none.

(h) Candidates.

Minimum companion mass (edge-on, $M_1 = 0.65 M_\odot$, from [equation \(6\)](#)) against orbital period, coloured by $\log D$; dotted lines mark the three companion-mass groups suggested for the Galactic bulge by Hajdu et al. [5]. This panel is descriptive: which of these are binaries is not established ([section 6](#)).

5.2 The cuts and why

1. $D > 40$. With 17,488 stars, fewer than one white-noise false alarm requires a per-star false-alarm probability below $\sim 6 \times 10^{-5}$, i.e. well beyond the largest null value ($D = 23$ in 2,998 simulations).

Table 2: Candidate cuts, applied in order. Columns: surviving real stars (all / with MACHO), and the fraction of each simulated class (MACHO campaign) still passing after the cut. LTTE: broad prior, $P_{\text{orb}} = 400\text{--}10^4$ d, $M_2 = 0.05\text{--}1.5 M_{\odot}$; emp.: empirical timing noise (section 4.8); r.w.: random walk.

	cut	real	LTTE	emp.	Blazhko	jump	large jump	large r.w.	null
1	$D > 40$	1009 / 812	0.304	0.105	0.316	0.062	0.809	0.212	0
2	$\chi^2_{\nu}(\alpha) < 2$	366 / 296	0.302	0.105	0.065	0.062	0.795	0.163	0
3	$A/\sigma_{\text{season}} > 3$	346 / 276	0.246	0.103	0.057	0.047	0.778	0.163	0
4	≥ 1.5 cycles	92 / 76	0.160	0.043	0.057	0.001	0.109	0.073	0
5	$A < a_1/c (M_2 = 2 M_{\odot})$	75 / 69	0.160	0.026	0.022	0.001	0.090	0.025	0

$D > 40$ is chosen conservatively high because the real contaminants are not white noise; lowering it mainly admits more intrinsic period changes (panel a).

- Amplitude constancy, $\chi^2_{\nu}(\alpha) < 2$.** Motivated by the physics: the LTTE is a pure time shift (figure 2), so any season-to-season amplitude change points to Blazhko modulation (or changing blends). The cut removes 79% of simulated Blazhko stars that passed cut 1 ($0.316 \rightarrow 0.065$) while keeping 99% of the orbits ($0.304 \rightarrow 0.302$), and 64% of real stars with $D > 40$. It removes *no* empirical-noise simulations, because that class has no amplitude modulation, whereas it removes most real high- D stars; this matters for the contamination estimate (section 6).
- $A/\sigma_{\text{season}} > 3$. The orbital amplitude must be large compared with a single season’s error, so that the curve is seen season by season and is not only an integrated excess; it mainly guards against low-amplitude, high-jitter solutions.
- At least 1.5 orbital cycles.** The decisive cut against intrinsic period changes: a sinusoid fitted over less than about one cycle mimics a smooth trend. It removes 86% of the large abrupt period changes ($0.778 \rightarrow 0.109$), 55% of the large random walks and 58% of the empirical noise, at the cost of a third of the simulated orbits: those with periods longer than about 2/3 of the baseline, which cannot be confirmed as periodic anyway.
- Physical ceiling.** The fitted amplitude must be achievable with $M_2 \leq 2 M_{\odot}$ at the fitted period (panel c). At this stage it removes no simulated orbits but 2/3 of the remaining large random walks ($0.073 \rightarrow 0.025$) and 17 real stars. The cut is applied identically to data and simulations.

The cuts were set on physical grounds and on the simulations, not tuned on the real candidates. They are not a clean binary selection: large abrupt period changes still pass at 9%, the empirical noise at 2.6% and Blazhko at 2.2% (table 2). How many of the real candidates are expected from noise alone is the subject of section 6.

6 Part B: what the population says

Part B runs the full pipeline on simulated stars (section 5) to measure, with the same cuts as for the data, how often each kind of signal passes, how complete the selection is for real orbits, and how many candidates the real stars’ own timing noise produces. Scripts: `scripts/partB_analysis.py` (`results/partB/summary.json`, `completeness.csv`), `scripts/partB_bkg_limit.py`, `scripts/plot_partB.py` (figure 7).

6.1 The empirical noise reproduces the real stars

Among MACHO stars, the fraction with D above 10, 20, 40, 80 is 0.752, 0.327, 0.123, 0.017 for the real stars and 0.783, 0.325, 0.105, 0.015 for the empirical-noise simulations; white noise gives 0.243, 0.002, 0, 0 (figure 7a). The whole high- D tail of the real population, including the region of the candidates, is therefore what the measured intrinsic timing noise produces. Pure noise (null) never passes the cuts.

6.2 How many candidates does noise alone produce?

The empirical class has no amplitude modulation, so it passes the amplitude veto (cut 2) almost always, whereas only 46% of the real stars that pass the other four cuts also pass it. Real timing noise often comes with Blazhko-like amplitude changes. The comparison is therefore made in two ways (figure 7c):

- **All cuts except the amplitude veto** (where the empirical class is a valid background model): 150 real MACHO stars against 171 ± 13 expected from noise (0.0259×6612 , Poisson error).
- **All cuts, the veto corrected** with the real pass fraction (0.46): 69 real candidates against $\approx 79 \pm 9$.
- **Tier-1 proxy** (all cuts, $\chi^2_\nu(\alpha) < 1.5$, coherence $\chi^2_\nu < 1.5$, predictive score > 2 ; the Keplerian-fit and crowding criteria of Part C cannot be applied to simulations), veto corrected with the real pass fraction of this stricter cut (0.33): 17 real against $\approx 24 \pm 5$.

In none of these is there an excess of orbit-like signals over the timing noise. The OGLE-only sample (10,876 stars) has 6 candidates against 14 expected (95% upper bound 31), and no star passes the Tier-1 proxy (OGLE-only stars have no predictive test).

Caveat (circularity). The empirical noise is drawn from the REML noise fits of the real stars (candidates excluded). Undetected orbits in the other stars are absorbed into those fits, so the noise model can contain some orbital signal. This makes the background estimate conservative for detection, but it means that the population comparison cannot by itself prove that *no* candidate is a binary. The veto correction assumes that binaries and noise pass the amplitude veto at the same rate in the real data; a planned refinement adds measured amplitude modulation to the empirical class so that the veto is calibrated directly.

6.3 Completeness

For injected orbits (MACHO stars, isotropic, half eccentric; figure 7d), the cuts recover $M_2 = 0.4\text{--}1.5 M_\odot$ companions at 11% (400–1000 d), 56% (1–3 kd) and 41% (3–10 kd); $M_2 = 0.15\text{--}0.4 M_\odot$ at 0.5%, 10% and 15%; and $M_2 < 0.15 M_\odot$ at $\leq 2\%$. For OGLE-only stars (baseline 1997/2001–2016) the completeness is far lower: 2%, 13% and 2% for $M_2 = 0.4\text{--}1.5 M_\odot$, because of the shorter baseline and the 1.5-cycle cut.

6.4 Upper limits on the binary fraction

For $M_2 = 0.4\text{--}1.5 M_\odot$ and $P_{\text{orb}} = 1\text{--}10$ kd (completeness 0.478 for MACHO stars):

- **Model-free** (every candidate treated as a binary; 95% Poisson upper limit on 69 is 84.3): $f < 84.3/(6612 \times 0.478) = 2.7\%$ for MACHO stars and $f < 1.5\%$ for OGLE-only stars (6 candidates, low completeness). For $M_2 = 0.15\text{--}0.4 M_\odot$: $f < 9.9\%$ (MACHO) and $< 11\%$ (OGLE-only).

- **Background-subtracted** (150 stars before the amplitude veto, background $b = 171$ with a Gaussian systematic uncertainty, flat prior on the signal, Poisson likelihood): $f < 1.4\%$, 2.4% and 4.1% for a 15%, 25% and 50% systematic on b . These depend on the noise model and are quoted as a range.

The binary fraction of LMC RRab with massive companions on 3–30-yr orbits is therefore at most a few per cent. **Context.** The rotating, disc-like RR Lyrae of the Milky Way behave like a young, metal-rich thin-disc population [9]; the most metal-rich ones ($[\text{Fe}/\text{H}] > -0.5$) have kinematic ages of $\sim 6\text{--}7$ Gyr [10]. Binary evolution can produce such stars: partial envelope stripping on the red-giant branch places them in the instability strip, and all metal-rich RR Lyrae are then predicted to have an A-, F-, G- or K-type companion with $P_{\text{orb}} > 1000$ d [8], typically $\sim 900\text{--}2000$ d [11]. Gaia DR3 astrometry shows some tension with a high hidden binary fraction [11]. Our search is most complete in this regime of period and companion mass (56% at 1–3 kd for $M_2 = 0.4\text{--}1.5 M_{\odot}$). The LMC RRab are, however, dominated by old, metal-poor stars, so these limits apply to the population as a whole and do not test the metal-rich binary channel directly.

Implication. Population statistics give a limit, not a detection. Whether any individual candidate is a binary can only be decided with more data: further orbital cycles (OGLE-IV after 2016, [section 8](#)), Gaia DR4 photometry ([section 7](#)) and radial velocities.

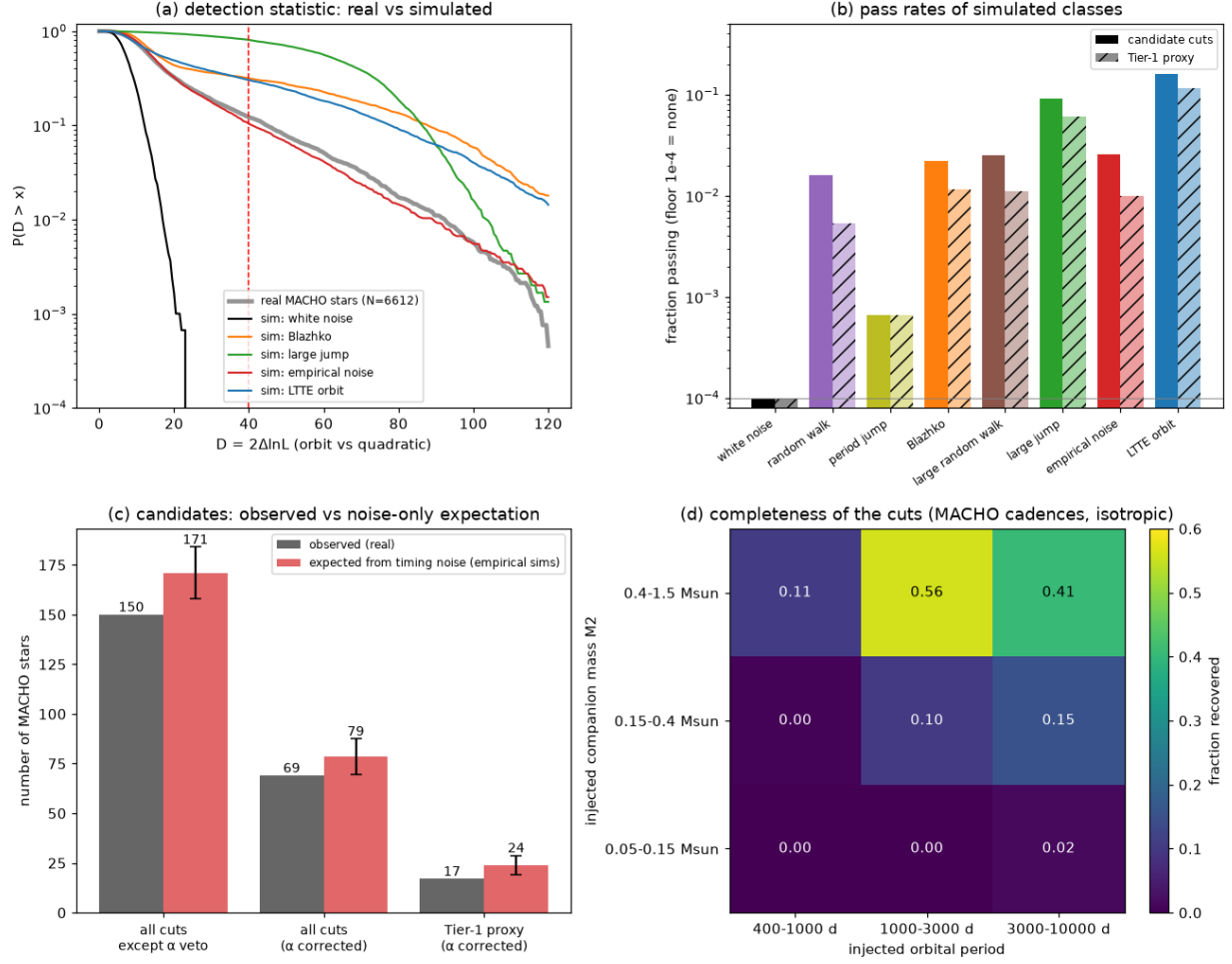


Figure 7: Part B, MACHO stars (6,612 real; 26,982 simulations). (a) Survival function of D : real stars (thick grey) against simulated classes; the empirical-noise class (red) follows the real distribution. (b) Fraction of each simulated class passing the candidate cuts (solid) and the Tier-1 proxy (hatched); white noise passes neither (plotted at the 10^{-4} floor). (c) Real candidates against the number expected from the empirical timing noise, at three stages; the amplitude veto is corrected with the real stars' pass fraction (see text); error bars are Poisson. (d) Completeness of the candidate cuts for injected LTTE orbits (isotropic, half eccentric) by orbital period and companion mass.

7 Part C: the candidates compared with the Galactic-bulge sample

This section uses the v3 pipeline (Part A of the refit: MACHO R , harmonic coherence, whole-cycle band alignment, band-lag priors, smooth-wander noise model); the contamination rates are those of Part B (section 6). Scripts: `scripts/partC_*.py`, `compare_hajdu.py`, `gaia_timing*.py`.

7.1 Our v3 candidates in brief

The cuts of table 2 applied to the v3 statistics select 75 stars (69 with MACHO). Each was fitted with a full Keplerian LTTE orbit (Irwin delay + quadratic + MACHO offsets with priors + white jitter; multi-start least squares, residual bootstrap; recovery of injected $e = 0, 0.4, 0.7$ orbits verified). Independent of the detection statistic we examined: the per-season amplitude constancy and harmonic coherence in every band, the out-of-sample prediction of the 1992–1996 MACHO seasons, crowding indicators from Gaia DR3 (MACHO candidates are indistinguishable from their parent population; KS $p = 0.2$ – 0.99), and the Keplerian fit quality. Grouping:

- **Tier 1** (11): Keplerian fit not at the eccentricity bound and covering ≥ 1.5 cycles, $\chi^2_\nu < 1.5$; amplitude and coherence $\chi^2_\nu < 1.5$; no crowding flag; predictive score > 2 . All have MACHO B , R and OGLE I . $P_{\text{orb}} = 5.4$ – 14.5 yr, $e = 0.15$ – 0.57 , $M_{2,\text{min}} = 0.14$ – $0.96 M_\odot$, $K_1 = 2.5$ – 11.5 km s^{-1} , $V = 19.2$ – 19.7 .
- **Tier 2** (17): at most one soft flag; predictive score > 0 or unavailable. **Tier 3** (47): the rest.

Diagnostic sheets for all 75 are in `plots/partC/sheets/`; figure 8 shows one Tier-1 example.

7.2 The bulge sample of Hajdu et al. (2021)

Hajdu et al. [5] examined $\sim 27,500$ OGLE-III/IV O–C diagrams of bulge RRab (data to 2019; mean/median baselines 10.5/7.4 yr), pre-selected ~ 400 by eye, and retained 87 after MCMC LTTE fits including a period-change term, in three quality classes (Q1: 25, P_{orb} uncertainty $< 1\%$; Q2: 32; Q3: 30, problematic fits). Their Table 1 was parsed from the arXiv source (`scripts/parse_hajdu2021.py`).

Follow-up status. We found no published confirmation or refutation of any of the 87. A 2026 study co-authored by Hajdu and Prudil states that “only a single RRL is confirmed as belonging to a binary system” [14], namely TU UMa, a field star [7]. Radial-velocity tests of Galactic field timing candidates have been largely negative (15 of 19 without binarity, Barnes et al. 15; KIC 2831097 ruled out, Poretti et al. 16). The status of the bulge candidates is therefore *untested*, not refuted. (Literature search: `docs/reviews/hajdu_followup.md`; an ADS full-text query could not be run.)

7.3 Side-by-side comparison

Where the bulge sample is stronger: precision. At $I \approx 15$ – 16 the bulge stars are 3–4 mag brighter; their orbital parameters are 2–4 times more precise (σ_A/A 1.7% vs 6.7%, σ_P/P 0.5% vs 1.1%, σ_e 0.02 vs 0.10 for Q1 vs Tier 1). Their seasons were also split into segments (more O–C points). Their posterior widths assume white timing noise beyond a constant $\dot{P}$, which understates the uncertainty if the period wanders (section 4.8).

What our sample adds, and what it does not. The main way an RR Lyrae imitates an orbit is intrinsic phase wander (irregular period changes are common in every population studied; e.g. Alcock et al. 17 for LMC RRc). Our Tier 1 differ from the bulge candidates as follows:

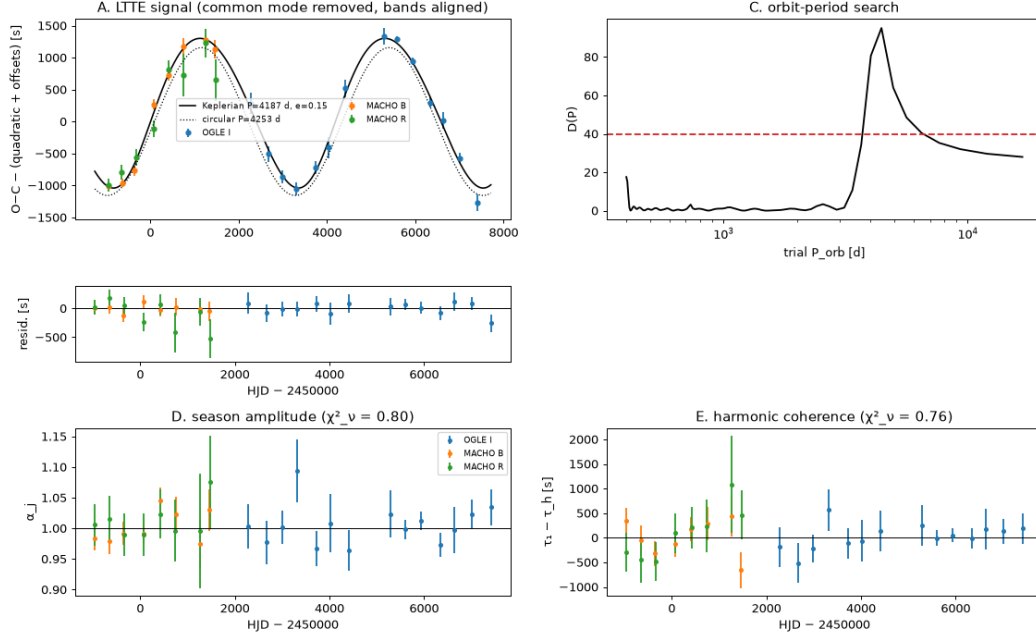


Figure 8: Candidate sheet for the Tier-1 star 10449. A: $O-C$ after removing the quadratic and the MACHO offsets (common mode removed, bands aligned), with the Keplerian ($P = 4187$ d, $e = 0.15$) and circular fits; OGLE I , MACHO B , MACHO R ; below, residuals. C: D against trial orbital period. D: per-season amplitude factors α_j by band. E: harmonic coherence by band. Both D and E are consistent with a pure time shift.

1. **Independent instruments** (MACHO B , R and OGLE I). This rules out *instrumental* artefacts (time stamps, photometric systematics), but *not* intrinsic wander, which every telescope sees identically (the same-season timing residuals of MACHO and OGLE are correlated, $\rho = 0.37$, section 4.8).
2. **More cycles** (median 2.9 vs 1.9). This helps, but a smooth wander with a correlation time of a few years still produces one or two bumps over 24 yr, which a sinusoid plus a quadratic fits.
3. **An out-of-sample prediction passed** (1992–1996 MACHO seasons). A partial safeguard only: smooth wander can continue smoothly into the held-out years (simulated wander passes the full Tier-1 proxy $\sim 1\%$ of the time).
4. **Calibrated contamination:** simulations with a noise model matched to the real stars. This is what shows that the candidate *number* is consistent with wander alone (Part B, section 6); the bulge analysis has no such calibration.

Where they are alike. Our sensitivity limit ($A \gtrsim 350$ s) restricts us to the high-amplitude regime of the bulge Q1 class: medians $A \approx 800$ s, $M_{2,\min} \approx 0.45\text{--}0.5 M_\odot$, $K_1 \approx 6$ km s $^{-1}$ in both (figure 9a, f). We cannot test the low-mass Q2/Q3 population ($M_{2,\min} \sim 0.1\text{--}0.2 M_\odot$).

Expectation. The v3 simulations (Part B) show that the number of our candidates is consistent with the tail of the real stars’ own timing noise (69 observed vs ≈ 79 expected from noise at the

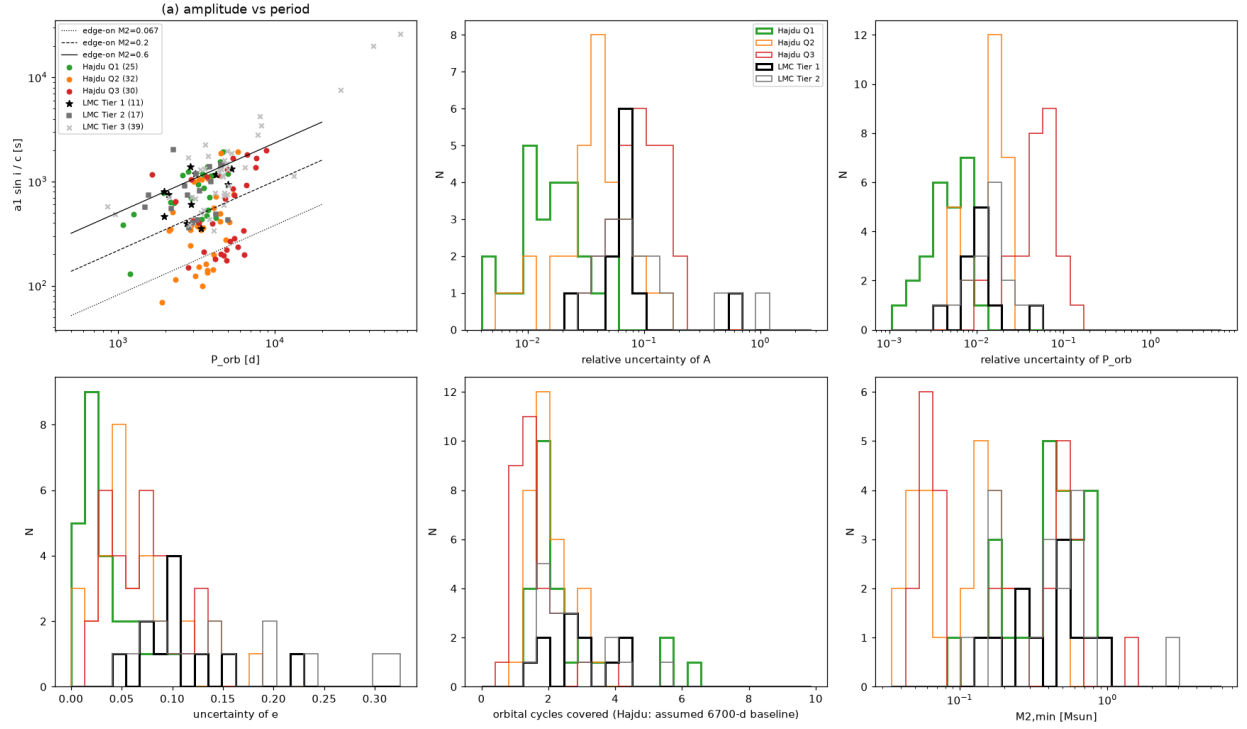


Figure 9: Our candidates (Tier 1: stars; Tier 2: squares; Tier 3: crosses) against the Hajdu Q1/Q2/Q3 bulge candidates: (a) LTTE amplitude vs orbital period with edge-on curves for $M_2 = 0.067, 0.2$ and $0.6 M_\odot$ (the three groups suggested by Hajdu et al.); (b, c) relative uncertainties of A and P_{orb} ; (d) uncertainty of e ; (e) orbital cycles covered; (f) minimum companion mass.

Table 3: Medians per class. $A = a_1 \sin i/c$; σ_A/A , σ_P/P , σ_e : Hajdu posterior standard deviations, ours bootstrap half 16–84% widths; cycles = baseline/ P_{orb} (Hajdu baseline *assumed* 6700 d, 2001.4–2019.7).

class	N	P_{orb} [d]	A [s]	e	$M_{2,\text{min}}$	K_1	σ_A/A	σ_P/P	σ_e	cycles
Hajdu Q1	25	3437	779	0.27	0.44	6.5	0.017	0.005	0.024	1.9
Hajdu Q2	32	3477	355	0.31	0.14	2.6	0.037	0.015	0.054	1.9
Hajdu Q3	30	4910	534	0.35	0.20	3.4	0.092	0.053	0.065	1.4
LMC Tier 1	11	2914	804	0.34	0.49	6.2	0.067	0.011	0.102	2.9
LMC Tier 2	17	3131	736	0.32	0.44	6.2	0.070	0.016	0.127	2.4

candidate cuts; 17 vs ≈ 24 for a Tier-1 proxy; [section 6](#)). Neither sample therefore establishes binarity statistically; the same is likely true of the bulge sample, whose selection was not calibrated against intrinsic wander. What separates an orbit from wander is strict repetition over more cycles, i.e. more data. Confirmation is, however, harder: at $V \approx 19.4$, measuring $K_1 \sim 6 \text{ km s}^{-1}$ requires $\sim 1\text{--}2 \text{ km s}^{-1}$ per epoch after removing a $\sim 60 \text{ km s}^{-1}$ pulsation curve (published LMC RR Lyrae RVs reach $10\text{--}35 \text{ km s}^{-1}$ per epoch).

7.4 A photometric test with Gaia

Gaia DR3 epoch G photometry (2014.6–2017.4) exists for 62 of the 75 candidates (median 34 usable transits, per-transit S/N ≈ 59). We fitted a G -band template and two delays, one per half of the DR3 window; the unknown G – I band lag cancels in their difference $\Delta\tau$, which is compared with each Keplerian prediction.

- **Calibration on 228 quiet non-candidate stars** (Gaia $\Delta\tau$ vs OGLE’s own $\Delta\tau$ at the same epochs, OGLE errors included): robust scatter of $z = 1.64$ (2.2% beyond 5σ) $\Rightarrow$ Gaia errors inflated by 1.64.
- **DR3 result:** only 4 of 53 testable candidates are informative ($|\Delta\tau_{\text{pred}}| > 2\sigma$). Two agree (16755, Tier 1, $z = 0.8$; 13469, $z = 0.3$); two disagree (13392, Tier 2, $z = -6.7$; 01106, Tier 3, $z = +3.5$), versus ~ 0.5 expected by chance from the control tail. DR3 is too sparse for most orbits.
- **DR4 forecast** (expected December 2026; 2014.6–2020.1, *assumed* $\sim 2\times$ the DR3 epochs; bins 2014.6–2017.4 and 2017.4–2020.1, the latter entirely after the public OGLE-IV data): testable at $> 2\sigma$ for 6 of 9 Tier 1 (5 at $> 3\sigma$), 8 of 13 Tier 2, 13 of 31 Tier 3. DR4 can therefore confirm or reject most of the strongest candidates photometrically, at no observing cost. OGLE-IV photometry after 2016 (not public) would be a stronger test still ([section 8](#)).

8 Frozen predictions and the OGLE data after 2016

8.1 OGLE kept observing the LMC

The public OCVS light curves of the original catalogue stars end at HJD’ ≈ 7516 (2016.4). OGLE-IV observed the LMC continuously until March 2020, resumed in August 2022 and is still operating; the catalogue extensions of 2017/2019 (1,432 stars) carry public data to 2020.2 and show 26–110 epochs per season in our candidates’ fields (`scripts/ogle_post2016_cadence.py`). A high-cadence survey of the central LMC fields ran from October 2022 to May 2024 [[13](#)]. Its field list (LMC502, 503, 509, 510, 516) is from an agent’s reading of the paper and has not been checked. These fields host 49 of the 75 candidates, including 7 of the 11 Tier-1 stars. At ~ 2400 epochs per season the expected

delay error is $132\text{ s} \times \sqrt{54/2400} \approx 20\text{ s}$ per season, against 132 s now. These data are available only on request from the OGLE team; a request for all LMC OCVS RR Lyrae has been drafted (`docs/ogle_request_email_short.md`).

8.2 What is frozen

Before any new data are seen, the predicted OGLE I delays for 2016.4–2026.5 have been computed for the 28 Tier-1/2 candidates from the public data alone (`scripts/freeze_predictions.py`, `src/rrlbin/predict.py`), under both hypotheses:

- H_1 : the best Keplerian orbit, plus the smooth-wander GP refitted to the orbit-subtracted delays;
- H_0 : no orbit, with the star’s own REML GP noise model ([section 4.8](#)).

Each is a universal-kriging prediction (GP conditional mean and covariance, including the uncertainty of the quadratic and MACHO offsets); the full covariances are stored. A residual bootstrap of the Keplerian fit (200 refits) is stored as well. Note that the H_1 kriging covariance is conditional on the best-fit orbit, so the bootstrap envelope should be added when new data are compared. The table `results/predictions/predictions_2026-10-01.parquet` has SHA-256 `a79bc756...607813d`, recorded with the code commit in `_meta.json`, and is tagged `predictions-2026-10-01` in the repository. The predictive standard deviations (medians) are 179 s (2020) and 282 s (2024) under H_1 , and 1.6 ks and 2.8 ks under H_0 ([figure 10](#)).

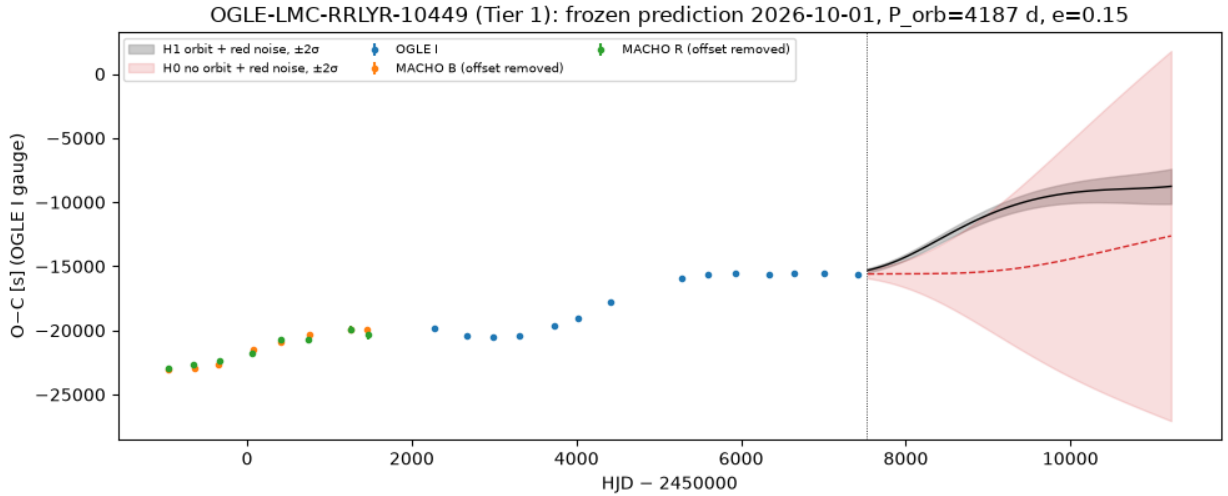


Figure 10: Frozen prediction for the Tier-1 candidate 10449 ($P_{\text{orb}} = 4187\text{ d}$, $e = 0.15$, $A = 1176\text{ s}$; its $O-C$ in the orbit frame is in [figure 8](#)). Points: public data in the OGLE I gauge (MACHO offsets removed). Grey: H_1 (orbit + red noise), $\pm 2\sigma$; red: H_0 (no orbit, red noise), $\pm 2\sigma$; dotted line: end of the public data.

8.3 How decisive will the new data be?

For new OGLE season delays (130 s error plus each star’s white jitter), the expected log Bayes factor is the Kullback–Leibler divergence between the two predictive distributions: $\langle \ln \text{BF} \rangle = \text{KL}(N_1 \| N_0)$ if the orbit is real and $-\text{KL}(N_0 \| N_1)$ if it is not (`scripts/predictions_decisiveness.py`). Medians:

	2017–2020 (3 seasons)		2017–2026 (7 seasons)	
	orbit real	no orbit	orbit real	no orbit
Tier 1 (11)	+4.0 (7/11 > 3)	−53	+8.4 (11/11 > 3)	−200
Tier 2 (17)			+7.2 (17/17 > 3)	−101

These numbers assume 130 s per season (the high-cadence fields would give ≈ 20 s) and the noise model of section 4.8. They are optimistic to the extent that the H_1 covariance omits the orbit-parameter uncertainty. Even so, the 2017–2026 OGLE data would confirm or reject every Tier-1/2 candidate. The asymmetry is expected: a wandering star almost never follows a specific narrow orbital prediction for a decade, whereas a true orbit is only moderately unlikely under the broad H_0 .

9 Summary and next steps

- We measured per-season pulsation delays for 17,488 LMC RRab (OGLE-II/III/IV 1997–2016; MACHO B , R 1992–1999 for 6,612) and searched the $O-C$ diagrams for light-travel-time orbits.
- 75 stars pass the candidate cuts (11 Tier 1, 17 Tier 2). The Tier-1 orbits have $P_{\text{orb}} = 5.4\text{--}14.5$ yr, $A = 350\text{--}1380$ s and $M_{2,\text{min}} = 0.14\text{--}0.96 M_{\odot}$.
- The real stars carry red, mostly intrinsic timing noise. Simulations with that noise reproduce the real D distribution and predict as many candidates as are observed: there is no population excess. The binary fraction for $M_2 = 0.4\text{--}1.5 M_{\odot}$, $P_{\text{orb}} = 1\text{--}10$ kd is below 2.7% (model-free; 1.4–4.1% background-subtracted).
- Individual candidates are untested, like the bulge candidates of Hajdu et al. [5]. Predictions for 2016–2026 are frozen; OGLE-IV data after 2016 would decide every Tier-1/2 candidate, and Gaia DR4 most Tier-1 ones.

Next steps. (1) Add amplitude modulation, drawn from the real stars’ α_j variability, to the empirical noise class, and re-run the Part B comparison with the amplitude veto calibrated directly. (2) Obtain the post-2016 OGLE-IV photometry and test the frozen predictions; re-run the full analysis on the 1992–2026 baseline. (3) Test the Tier-1 candidates with Gaia DR4 epoch photometry. Season-level outlier rejection in Level 2 is a minor open item (e.g. one bad MACHO R season of 13854).

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